

PARIKSHIT MIT — TASK 2

MINIATURIZED SATELLITE FORMATS

{Used in Space Engineering & Education}

Name : B SaiGanesh

Subsystem : Admin & Operations(AO)

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INTRODUCTION

Satellites are important tools used for communication, navigation, weather monitoring, Earth observation, scientific research and many other applications.

Miniaturized satellite formats provide a simpler and more accessible way to learn and develop space technology. They reduce the size, mass and cost of the system while still allowing engineers to work with important spacecraft technologies such as sensors, computers, communication systems, power systems.

CubeSats are standardized small spacecraft that can operate in space.

CanSat, Educational systems that simulate functions of a satellite without entering orbit.

FlatSats are ground-based setups used to develop, integrate satellite subsystems before they are placed inside the actual spacecraft.

This report studies CubeSat, CanSat and FlatSat in detail. It also introduces other educational satellite formats such as PocketQube, BalloonSat and TubeSat. The report compares these formats and explains their importance in space engineering education, practical learning and technology development.

MINIATURIZED SATELLITES

Miniaturized satellites are small and lightweight systems used for space-related experiments, education and research. They are designed to reduce the cost and complexity of satellite projects.

They help students learn practical concepts such as electronics, programming, sensors, communication, power systems and data collection.

Some common miniaturized satellite formats are CubeSat, CanSat and FlatSat. Each has a different purpose and is used at a different stage of learning or satellite development.

CUBESAT

A CubeSat is a small standardized satellite used for education, research and technology development. Its size is based on a standard unit called **1U (one unit)**.

1U CubeSat = $10 \times 10 \times 10$ cm. Multiple units can be combined. Examples 2U, 3U and 6U.

Main Components

A CubeSat contains different subsystems that work together. These include power, communication, onboard computing, attitude control, structure and thermal management, and the payload.

Educational Importance & Engineers Purpose

CubeSats give students practical experience in designing, building and testing a real spacecraft. They help students understand how different engineering systems work together in a space mission.

A CubeSat is made up of different subsystems.

- **ADCS** – Attitude Determination and Control System:
Determines and controls the orientation of the satellite.
- **COMMS** – Communication System:
Sends data from the satellite to the ground station and receives commands.
- **EPS** – Electrical and Power System:
Generates, stores and distributes electrical power.
- **ODHS** – OnBoard Data Handling System:
Processes and manages data inside the satellite.
- **STMS** – Structural and Thermal Management System:
Provides physical support and helps control the satellite's temperature.
- **Payload**: Performs the main experiment or mission-specific task of the satellite.
- **Admin & Operations**: Supports planning, documentation, coordination and mission operations.

CANSAT

A CanSat is a small educational system designed to simulate some functions of a real satellite.

A CanSat can contain sensors, a microcontroller, battery and communication equipment can fit inside a container similar in size to a soft-drink can.

It is usually carried to a certain altitude using a rocket or balloon and collects data during its flight.

Difference from CubeSat, as CubeSat is a small spacecraft that can operate in space, while a CanSat is mainly an educational simulation and normally does not enter orbit.

Educational Use

CanSat projects help students learn about sensors, programming, electronics, communication and data collection. They also give students practical experience in planning and conducting a small mission.

FLATSAT

A FlatSat is a ground-based setup used to develop and test a satellite. The main satellite electronics and subsystems are arranged on a flat surface instead of being installed inside the final satellite structure.

FlatSats are used to test whether different satellite subsystems work correctly together.

Educational Use

They help engineers find problems and make changes before the actual satellites are assembled and launched. Students can use FlatSats to understand satellite electronics, communication, power systems, sensors and data handling in a practical way.

Difference from CubeSat as, a CubeSat is an actual spacecraft that can be launched into space, while a FlatSat is a ground-based testing and development setup.

OTHER EDUCATIONAL SATELLITE FORMATS

1. Pocket Qube

PocketQube is a very small satellite format designed for small experiments, research and educational projects. Its smaller size allows students to work on simple and low-cost satellite missions.

2. BalloonSat

A BalloonSat is a small educational payload carried by a high-altitude balloon. It can carry sensors, cameras and other instruments to collect data at high altitude. It helps students learn about space-related experiments without sending a satellite into orbit.

3. TubeSat

TubeSat is a small satellite format with a tube-shaped design. It can be used for educational projects and technology experiments. Its simple design can be used for learning about small spacecraft and technology experiments.

COMPARISON OF DIFFERENT FORMATS

Format	Basic Idea	Main Use
FlatSat	Ground-based satellite testing setup	Testing and development
BalloonSat	Payload carried by a balloon	High-altitude experiments
CanSat	Satellite simulation in a small container	Student experiments and learning
CubeSat	Small standardized spacecraft	Education, research and space missions
TubeSat	Small tube-shaped satellite	Education and technology experiments
PocketQube	Very small satellite	Small experiments and education

The main difference is that a CubeSat is designed to operate as a spacecraft, while a CanSat is mainly used for educational simulation and a FlatSat is used for ground testing.

EDUCATIONAL AND ENGINEERING IMPORTANCE

Miniaturized satellite projects provide students with practical experience in space engineering. Instead of learning only from textbooks, students can design, build, test and operate small satellite systems.

- **Electronics and sensors** – measuring and collecting physical data.
- **Programming** – controlling systems and processing collected data.
- **Communication** – sending information between the satellite and ground station.
- **Power systems** – generating, storing and distributing electrical power.
- **Structures and thermal systems** – protecting components and managing temperature.
- **Data handling** – collecting, storing and processing information.
- **Testing and troubleshooting** – finding and fixing problems before a mission.
- **Teamwork and project management** – working together on different parts of a mission.

For engineers, small satellite platforms can also be used to test new technologies at a smaller scale.

Overall, miniaturized satellite projects connect classroom knowledge with real engineering practice and help students develop skills needed for future space missions.

PARIKSHIT MIT AND CUBESAT

PARIKSHIT MIT is a student satellite project at Manipal Institute of Technology that focuses on developing a small spacecraft using the **CubeSat format**. It gives students an opportunity to apply their classroom knowledge to a practical space engineering project.

PARIKSHIT is based on a **2U CubeSat**, where 2U means that the satellite uses two standard CubeSat units. The satellite is designed to work as one complete system, with different subsystems responsible for different functions.

The project also involves activities such as system design, testing, integration, documentation and mission operations. Students from different engineering backgrounds can contribute to different parts of the project and learn how a real satellite mission is developed.

Therefore, **PARIKSHIT** is a good example of how the CubeSat format can be used for hands-on education, teamwork and practical space engineering experience.

ADVANTAGES AND LIMITATIONS

Advantages

- **Low cost:** Smaller systems generally require fewer resources to develop.

- **Small and lightweight:** They are easier to handle, test and transport.
- **Faster development:** Projects can often be designed and developed in less time.
- **Practical learning:** Students gain hands-on experience with real engineering concepts.
- **Technology development:** New ideas and technologies can be tested on a smaller scale.

Limitations

- **Limited space:** Small satellites have less room for equipment and components.
- **Limited power:** They have less space for power generation and storage.
- **Limited capability:** They may not perform complex missions possible with larger satellites.
- **Communication limitations:** Small systems may have restrictions on communication range and data capacity.
- **Mission limitations:** Formats such as CanSat and FlatSat are mainly used for education, testing and development rather than orbital missions.

CONCLUSION

Miniaturized satellite formats provide a practical and affordable way to learn and develop space technology. CubeSats can be used as real small spacecraft, while CanSats provide a simpler way to simulate satellite missions. FlatSats help students and engineers test satellite systems on the ground.

Other formats such as PocketQube, BalloonSat and TubeSat also provide opportunities for education and small-scale experiments.

Overall, miniaturized satellite projects connect theoretical knowledge with practical engineering and provide valuable experience for future space missions.

REFERENCES

1. **European Space Agency** (ESA) – Educational satellite and CubeSat resources.
2. **NASA** – Small satellite and CubeSat information.
3. **CubeSat.org** – CubeSat standards and information.
4. **NASA** – CanSat and student satellite educational resources.
5. **Others** –Educational and technical resources on PocketQube, BalloonSat and TubeSat.

Thank You PARIKSHIT MIT.